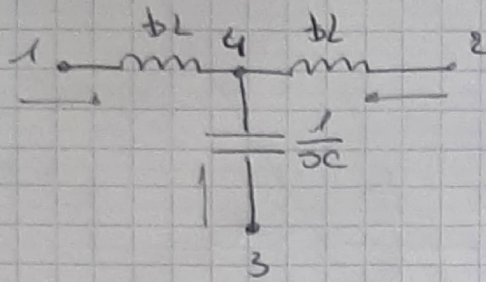


TS 7

① Construir la MAI.



$$MAI = \begin{pmatrix} Y_{11} & Y_{12} & Y_{13} & Y_{14} \\ Y_{21} & Y_{22} & Y_{23} & Y_{24} \\ Y_{31} & Y_{32} & Y_{33} & Y_{34} \\ Y_{41} & Y_{42} & Y_{43} & Y_{44} \end{pmatrix}$$

$$Y_{ij} = \frac{I_1}{V_i} \bigg|_{V_k=0} \quad V_{k+1}$$

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{3L} & 0 & 0 & -\frac{1}{3L} \\ 0 & \frac{1}{3L} & 0 & -\frac{1}{3L} \\ 0 & 0 & c & -c \\ -\frac{1}{3L} & -\frac{1}{3L} & -c & \frac{2}{3L} + c \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{pmatrix}$$

MAI

$$I_1 = V_1 \cdot \frac{1}{3L} + 0V_2 + 0V_3 + 0 - \frac{1}{3L} V_4$$

$$I_2 = V_1 \cdot 0 + V_2 \frac{1}{3L} + 0V_3 - \frac{1}{3L} V_4$$

$$I_3 = V_3 c - V_4 c$$

$$I_4 = V_4 \cdot \left(\frac{1}{3L} + \frac{1}{3L} + c \right) - V_2 \frac{1}{3L} - V_1 \frac{1}{3L} - cV_3$$

② $\frac{V_{23}}{V_{13}} = \frac{Y_{13} c}{Y_{13} c} = \frac{\frac{1}{3L} - \frac{1}{3L}}{\frac{1}{3L} - \frac{1}{3L}} = \frac{1}{\frac{2}{3L} + c} - \frac{1}{3L}$

$$\textcircled{2} \quad a) \quad \frac{V_{23}}{V_{13}} = \frac{V_{ij}}{V_{mn}} = \text{sgn}(m-n) \text{sgn}(i-j) \frac{Y_{ij}^{mn}}{Y_{mn}^{mn}} \quad \begin{matrix} mn \\ i+j \\ (-1)^{m+n} \end{matrix}$$

$$\frac{V_{23}}{V_{13}} = \frac{\text{sgn}(1-3) \text{sgn}(2-3)}{+1} \frac{\begin{pmatrix} 0 & -\frac{1}{\phi L} \\ -\frac{1}{\phi L} & \frac{2}{\phi L} + \phi c \end{pmatrix}}{\begin{pmatrix} \frac{1}{\phi L} & -\frac{1}{\phi L} \\ -\frac{1}{\phi L} & \frac{2}{\phi L} + \phi c \end{pmatrix}} \frac{-1}{1}$$

$$= \frac{+ \frac{1}{\phi^2 L^2}}{\frac{2}{\phi^2 L^2} + \frac{c}{L} - \frac{1}{\phi^2 L^2}} = \frac{\frac{1}{\phi^2 L^2}}{\frac{1}{\phi^2 L^2} + \frac{c}{L}} = \frac{1}{1 + \phi^2 L c}$$

$$\boxed{\frac{V_{23}}{V_{13}} = \frac{1/Lc}{\phi^2 + \frac{1}{Lc}}}$$

$$\frac{V_{21}}{V_{31}} = \frac{\text{sgn}(3-1) \text{sgn}(2-1)}{+1} \frac{Y_{21}^{31}}{Y_{31}^{31}} \frac{(-1)^2}{(-1)^8}$$

$$\frac{V_{21}}{V_{31}} = \frac{\begin{pmatrix} 0 & -\frac{1}{\phi L} \\ -\phi c & \frac{2}{\phi L} + \phi c \end{pmatrix}}{\begin{pmatrix} \frac{1}{\phi L} & -\frac{1}{\phi L} \\ -\frac{1}{\phi L} & \frac{2}{\phi L} + \phi c \end{pmatrix}} \frac{-1}{1} = \frac{\frac{c}{L}}{\frac{2}{\phi^2 L^2} + \frac{c}{L} - \frac{1}{\phi^2 L^2}}$$

$$\frac{V_{21}}{V_{31}} = \frac{\frac{c}{L}}{\frac{c}{L} + \frac{1}{\phi^2 L^2}} = \frac{\phi^2 L^2 \frac{c}{L}}{\phi^2 L^2 \frac{c}{L} + 1} = \frac{\phi^2 L c}{\phi^2 L c + 1}$$

$$\boxed{\frac{V_{21}}{V_{31}} = \frac{\phi^2}{\phi^2 + \frac{1}{Lc}}}$$

$$\frac{V_{12}}{V_{32}} = \frac{\text{sgn}(3-2) \text{sgn}(1-2)}{1}$$

$$\frac{Y_{32}}{Y_{32}} = \frac{(-1)^{3+2+1+2}}{(-1)^{3+2+3+2}}$$

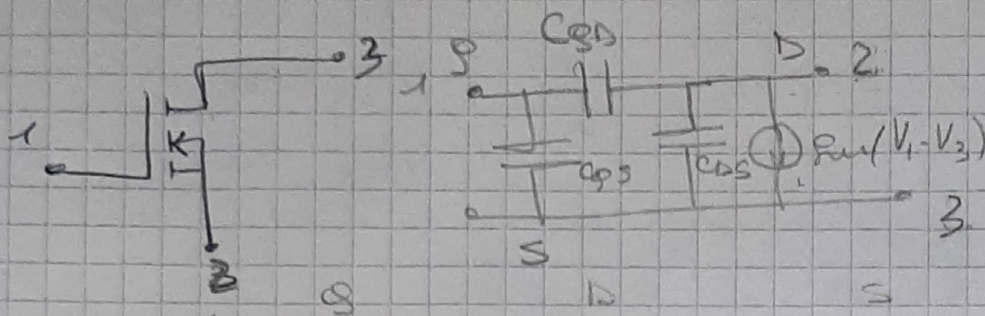
$$\frac{V_{12}}{V_{32}} = -1 \cdot \frac{\begin{pmatrix} 0 & -\frac{1}{\partial L} \\ -\partial c & \frac{2}{\partial L} + \partial c \end{pmatrix}}{\begin{pmatrix} \frac{1}{\partial L} & -\frac{1}{\partial L} \\ -\frac{1}{\partial L} & \frac{2}{\partial L} + \partial c \end{pmatrix}} = \frac{\frac{2}{\partial^2 L^2} + \frac{c}{L} - \frac{\partial}{\partial^2 L^2}}{\frac{2}{\partial^2 L^2} + \frac{c}{L} - \frac{\partial}{\partial^2 L^2}}$$

$$\frac{V_{12}}{V_{32}} = \frac{\frac{1}{\partial^2 L^2}}{\frac{1}{\partial^2 L^2} + \frac{c}{L}} = \frac{1}{1 + \frac{\partial^2 L^2 c}{L}}$$

$$\frac{V_{12}}{V_{32}} = \frac{\frac{1}{\partial^2 L^2}}{\frac{c}{L} + \frac{1}{\partial^2 L^2}} = \frac{\frac{L}{\partial^2 L^2}}{\frac{c}{L} + \frac{1}{\partial^2 L^2}}$$

$$\frac{V_{12}}{V_{32}} = \frac{\partial^2 c/L}{\frac{c}{L} + \frac{1}{\partial^2 L^2}} = \frac{c/L}{\frac{c}{L} + \frac{1}{\partial^2 L^2}}$$

$$\frac{V_{12}}{V_{32}} = \frac{\partial^2}{\partial^2 + \frac{1}{Lc}}$$



$$MII = \begin{pmatrix} s(C_{gs} + C_{gd}) & -sC_{gd} & -sC_{gs} \\ -sC_{gd} + g_m & s(C_{gs} + C_{ds}) & -g_m - sC_{ds} \\ -g_m - sC_{gs} & -sC_{ds} & g_m + s(C_{ds} + C_{gs}) \end{pmatrix}$$

$$C_{iss} = C_{gs} + C_{gd}$$

$$C_{oss} = C_{ds} + C_{gd}$$

$$C_{rss} = C_{gs}$$

$$C_{gs} = C_{iss} - C_{rss}$$

$$C_{ds} = C_{oss} - C_{rss}$$

$$MII = \begin{pmatrix} sC_{iss} & -sC_{rss} & s(C_{rss} - C_{iss}) \\ g_m - sC_{rss} & sC_{oss} & s(C_{rss} - C_{oss}) - g_m \\ s(C_{rss} - C_{iss}) - g_m & s(C_{rss} - C_{oss}) & g_m + s(C_{iss} + C_{oss} - 2C_{rss}) \end{pmatrix}$$

3) a)

$$Y_{sc} = \begin{pmatrix} sC_{iss} & -sC_{rss} \\ g_m - sC_{rss} & sC_{oss} \end{pmatrix}$$

$$Y_{dc} = \begin{pmatrix} sC_{iss} & s(C_{rss} - C_{iss}) \\ s(C_{rss} - C_{iss}) - g_m & g_m + s(C_{iss} + C_{oss} - 2C_{rss}) \end{pmatrix}$$

$$T_{SC} = \begin{pmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ \frac{Y_{12}Y_{21} - Y_{11}Y_{22}}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{pmatrix} \quad \begin{aligned} Y_{22} &= b_{CROSS} \\ Y_{21} &= q_m - b_{CROSS} \\ Y_{12} &= -b_{CROSS} \\ Y_{11} &= b_{CROSS} \end{aligned}$$

$$T_{SC} = \begin{pmatrix} -\frac{b_{CROSS}}{q_m - b_{CROSS}} & -\frac{1}{q_m - b_{CROSS}} \\ \frac{\Delta Y}{q_m - b_{CROSS}} & -\frac{b_{CROSS}}{q_m - b_{CROSS}} \end{pmatrix}$$

$$\Delta Y = -b_{CROSS} q_m + b_{CROSS}^2 - b_{CROSS}^2$$